

1. 由 Hopkinson's law

$$\begin{aligned}\widehat{\Phi}^{\text{FEM}} &= \frac{N_c}{R_a} \widehat{\mathbf{K}}_I \cdot \mathbf{I} = \frac{N_c}{R_a} \frac{6}{5} k_{11} \widehat{\mathbf{K}}_I^{\text{FEM}} \cdot \mathbf{I} \\ \widehat{g}_B &= \frac{6}{5} \frac{N_c k_{11}}{4\pi R_a \widehat{\ell}^2} \quad \Rightarrow \quad R_a = \frac{6}{5} \frac{N_c k_{11}}{4\pi \widehat{g}_B \widehat{\ell}^2} \\ \Rightarrow \widehat{\Phi}^{\text{FEM}} &= \frac{5}{6} \frac{4\pi \widehat{g}_B \widehat{\ell}^2}{N_c k_{11}} N_c \frac{6}{5} \widehat{\mathbf{K}}_I^{\text{FEM}} \cdot \mathbf{I} = \frac{4\pi \widehat{g}_B \widehat{\ell}^2}{k_{11}} \widehat{\mathbf{K}}_I^{\text{FEM}} \quad (\text{解 } k_{11})\end{aligned}$$

2. 再由

$$R_a = \frac{6}{5} \frac{N_c k_{11}}{4\pi \widehat{g}_B \widehat{\ell}^2} \quad (\text{解 } R_a)$$